

ORIGINAL ARTICLE

WILEY

Henagliflozin as add-on therapy to metformin in patients with type 2 diabetes inadequately controlled with metformin: A multicentre, randomized, double-blind, placebo-controlled, phase 3 trial

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Abstract

Aim: To evaluate the efficacy and safety of henagliflozin in patients with type 2 diabetes mellitus (T2DM) inadequately controlled with metformin.

Material and Methods: This multicentre phase 3 trial included a 24-week randomized, double-blind, placebo-controlled period, followed by a 28-week extension period. Patients with a glycated haemoglobin (HbA1c) level of 7.0% (53 mmol/mol) to 10.5% (91 mmol/mol) were randomized and treated with once-daily placebo ($n = 161$), henagliflozin 5 mg ($n = 162$), or henagliflozin 10 mg ($n = 160$). After 24 weeks, patients on placebo were switched to 5 mg or 10 mg henagliflozin for the additional 28-week treatment, and patients on henagliflozin during 24-week treatment period maintained this initial therapy. The primary endpoint was change in HbA1c from baseline to Week 24.

Results: At Week 24, the least squares mean HbA1c changes versus placebo from baseline were -0.76% (-8.3 mmol/mol) and -0.80% (-8.7 mmol/mol) for henagliflozin 5 and 10 mg, respectively (all $P < 0.0001$). Compared with the placebo group, both doses of henagliflozin lowered fasting plasma glucose, 2-hour postprandial plasma glucose, body weight and blood pressure, and increased the proportions of patients achieving HbA1c $< 7.0\%$ (53 mmol/mol) at Week 24. The trends in these improvements were sustained over an additional 28 weeks. Slightly higher proportions of ketosis and presence of urine ketone bodies were observed in patients treated with henagliflozin compared to placebo at Week 24. No diabetic ketoacidosis or episodes of severe hypoglycaemia were reported.

Conclusions: Henagliflozin 5 mg or 10 mg as add-on therapy to metformin provided a new therapeutic option for the treatment of T2DM patients who have inadequate glycaemic control with metformin alone, and was generally well tolerated.

Funding information

This study was funded by Jiangsu Hengrui Medicine.

KEYWORDS

add-on therapy, henagliflozin, metformin, sodium-glucose cotransporter type-2 inhibitor, type 2 diabetes mellitus

1 | INTRODUCTION

Metformin combined with lifestyle modification (diet and exercise) is still considered the first-line pharmacological treatment for the management of type 2 diabetes mellitus (T2DM).¹ If individualized glycosylated haemoglobin (HbA1c) targets are not achieved with metformin monotherapy in T2DM patients, other oral antidiabetic drugs (OADs) can be added to metformin, however, the benefits derived from lowering of HbA1c levels by the traditional OADs may be offset by risks of secondary failure, such as body weight gain.^{2–4}

Sodium-glucose cotransporter-2 (SGLT2) inhibitors, a relatively new class of OADs, specifically block glucose reabsorption in the proximal tubules of the kidney, which leads to urinary glucose excretion, and consequently, reduces blood glucose levels in patients with T2DM.^{2,5} Due to their insulin-independent mechanism of action, SGLT2 inhibitors have complementary therapy potential and can be combined with metformin for patients with T2DM.⁵ Dapagliflozin, canagliflozin, ertugliflozin and empagliflozin have been approved for the treatment of adults with T2DM since 2013, and these are widely considered as options for add-on therapy to metformin.⁶ The additional effects of SGLT2 inhibitors (body weight loss and blood pressure lowering) may contribute to the class effects of metabolic benefits, cardiovascular and renal protection.^{5,7,8} Currently, 2019 European Society of Cardiology/European Association for the Study of Diabetes guidelines and 2020 American Diabetes Association guidelines recommend prescribing an SGLT2 inhibitor following initial trial of lifestyle modifications and metformin in patients with chronic kidney disease, cardiovascular disease and heart failure.^{9,10}

Henagliflozin is an orally active, highly selective inhibitor of SGLT2 with more than 800-fold higher selectivity for SGLT2 compared to SGLT1. A phase 3 trial (NCT03159052) has demonstrated that henagliflozin monotherapy (5 or 10 mg once daily) provides efficacious glycaemic control, reduces body weight and blood pressure, and is generally well tolerated in patients with T2DM and inadequate glycaemic control despite diet and exercise.¹¹ The present study aimed to assess the efficacy and safety of henagliflozin when added to metformin in adults with T2DM who were not adequately controlled with metformin alone, and the results from the 24-week placebo-controlled period and the additional 28-week extension period are reported.

2 | MATERIALS AND METHODS

2.1 | Study design

This multicentre phase 3 study (NCT04390295) of henagliflozin in patients with T2DM who had inadequate glycaemic control with

metformin, conducted at 47 sites in China, included a randomized, double-blind, placebo-controlled, 24-week treatment period, followed by a 28-week extension period during which patients who received placebo and completed the 24-week treatment period switched to receive henagliflozin (Figure S1 in Appendix S1). The study complied with the Declaration of Helsinki and Good Clinical Practice guidelines, and was approved by institutional review boards and the independent ethics committees of the participating centres. All patients provided written, informed consent.

2.2 | Patients

Patients were eligible for inclusion if they were aged 18 to 75 years, were taking a stable dose of metformin (≥ 1500 mg/d) for at least 8 weeks before enrolment, and had T2DM, an HbA1c level of 7.5% (58 mmol/mol) to 11.0% (97 mmol/mol), as measured by local participating centres at screening, and 7.0% (53 mmol/mol) to 10.5% (91 mmol/mol), as measured by the central laboratory at randomization, a fasting plasma glucose (FPG) level ≤ 15 mmol/L, as measured by both participating centres at screening and the central laboratory at randomization, and a body mass index (BMI) of 19 to 35 kg/m². Key exclusion criteria were type 1 diabetes, cardiovascular disease (including New York Heart Association class III or IV congestive heart failure, unstable angina, myocardial infarction, severe cardiac arrhythmia or cardiac surgery), diabetic ketoacidosis within 6 months, urinary tract infection (UTI) and/or genital infection, renal impairment (defined as estimated glomerular filtration rate < 60 mL/min/1.73m²), systolic blood pressure (SBP) ≥ 160 mmHg or diastolic blood pressure (DBP) ≥ 100 mmHg.

2.3 | Randomization and study treatments

After a screening period of less than 3 weeks, patients received placebo to assess compliance with treatment during a 6-week, single-blind, lead-in period, after which, patients with enough compliance (between 80% and 120%), HbA1c between 7.0% (53 mmol/mol) and 10.5% (91 mmol/mol) and FPG ≤ 15 mmol/L measured by the central laboratory were randomized (in a 1:1:1 ratio) to receive double-blind henagliflozin 5 mg, henagliflozin 10 mg, or placebo, stratified by baseline HbA1c ($\leq 8.5\%$ [69 mmol/mol] or $> 8.5\%$ [69 mmol/mol]), for a 24-week treatment period, based on a computer-generated randomization code generated by an interactive web response system.

Patients in the placebo group were switched, in a 1:1 ratio, based on randomization conducted at the time of the 24-week treatment allocation, to receive henagliflozin 5 or 10 mg during the additional 28-week treatment after completing an initial 24-week double-blind treatment, and patients who received henagliflozin in the 24-week treatment period continued the same henagliflozin dose in the extension period. Regardless of mealtimes, henagliflozin 5 mg or 10 mg, and placebo were administered orally once daily as tablets in the morning. Then, all patients underwent a 4-week telephone follow-up period. During lead-in and throughout the duration of the study, patients received background metformin treatment (1500 mg/d) with lifestyle (diet and exercise) modification. Patients, investigators, evaluators and data analysts remained blinded to the treatment assignment from the time of randomization until the database was locked.

Glycaemic measurements were performed to determine the requirements for open-label rescue medication (any other oral hypoglycaemic drugs) for FPG concentrations more than 15.0 mmol/L during weeks 1 to 6, 13.3 mmol/L during weeks 7 to 12, or 11.1 mmol/L during weeks 13 to 52.

2.4 | Endpoints and assessments

The primary endpoint was change in HbA1c from baseline to Week 24, measured by the central laboratory. **Secondary efficacy endpoints** included changes from baseline in FPG, 2-hour postprandial plasma glucose (PPG), PPG area under the concentration-time curve 0 to 2 hours (AUC_{0-2h}), body weight, SBP and DBP at Week 24 and Week 52, change from baseline in HbA1c at Week 52, the proportion of patients achieving HbA1c <6.5% (48 mmol/mol) at Week 24 and the proportion of patients with HbA1c <7.0% (53 mmol/mol) at Week 24. Safety was evaluated based on reported adverse events (AEs), vital signs, physical examination, electrocardiographic findings as well as changes in clinical laboratory variables. Hypoglycaemia was defined as episodes with a glucose level $\leq$ 3.9 mmol/L (70 mg/dL) with or without symptoms.

2.5 | Statistical analysis

Efficacy was assessed in the full analysis set (FAS), including all randomized patients who received at least one dose of study treatment, and who had both a baseline and at least one post-baseline efficacy assessment during the 24-week treatment period or 28-week extension period. The safety analysis set (SAS) included all patients who received at least one dose of the study treatment and had at least once safety assessment during the 24-week period or 28-week extension period. For rescued patients, measurements obtained after initiation of rescue medication were not included in the efficacy analyses, but were included in the safety analyses.

We calculated that 360 patients (ie, 120 in the placebo, henagliflozin 5 mg, and henagliflozin 10 mg groups, respectively)

would provide 97% power to detect a reduction from baseline in HbA1c at Week 24 of 0.55% (6.0 mmol/mol) in the henagliflozin 10 mg group compared with the placebo group and 94% power to detect a reduction of 0.50% (5.5 mmol/mol) in the henagliflozin 5 mg group compared with the placebo group based on a *t*-test at a two-sided type I error of 0.05, on the assumption of a standard deviation (SD) of 1.1% (12 mmol/mol). An overall power of 91% was ensured for this study. For the primary endpoint, a hierarchical testing procedure was employed to strictly control the two-sided type I error at 0.05. The superiority of henagliflozin versus placebo was tested using a fixed-sequence stepdown approach from high dose to low dose. If the superiority of henagliflozin 10 mg was established, a further test for henagliflozin 5 mg would be performed. Multiple comparison procedures with hierarchical test procedures were not conducted for the secondary efficacy endpoints. Assuming a drop-out rate of 20%, 150 patients were required for each group and the total sample size was 450.

Treatment differences for the primary endpoint were assessed using analysis of covariance (ANCOVA) models, with treatment group as a fixed effect and baseline HbA1c as a covariate, and the missing data were imputed using the last observation carried forward (LOCF) method. The same ANCOVA models were applied for FPG, fasting body weight, SBP and DBP at Week 24 using LOCF methods, and 2-hour PPG and AUC_{0-2h} at Week 24 using observed case methods. Least squares (LS) mean changes from baseline, with 95% confidence interval (CI), were calculated for each group, and the LS mean difference and corresponding 95% CIs between treatment group and placebo were also evaluated, with *P* values for pairwise comparisons. Prespecified subgroup analyses by baseline HbA1c ($\leq$ 8.5% [69 mmol/mol] or $>$ 8.5% [69 mmol/mol]), sex (men or women), age (18-65 or $>$ 65 years), baseline BMI ($\geq$ 24 kg/m² vs. $<$ 24 kg/m²), and duration of type 2 diabetes ($<$ median or $\geq$ median) for the primary endpoint were also performed. Changes from baseline in HbA1c over 24 weeks were analysed by means of sensitivity analyses using a restricted maximum likelihood-based mixed model repeated measures approach with baseline HbA1c as linear covariates, and treatment, visit, and visit by treatment interaction as fixed effects, based on observed cases. Ad hoc analyses were conducted to further explore changes from baseline in HbA1c at Week 24, not excluding values after initiation of rescue therapy.

The proportions of patients achieving HbA1c <6.5% (48 mmol/mol) or HbA1c <7.0% (53 mmol/mol) at Week 24 were analysed with nonresponder imputation (patients with missing HbA1c value at Week 24 were considered nonresponders), and 95% CIs were computed by the Clopper-Pearson method. The proportions of patients achieving HbA1c <6.5% (48 mmol/mol) or HbA1c <7.0% (53 mmol/mol) at Week 24 were also compared between treatment and placebo groups using a Cochran-Mantel-Haenszel test, with baseline HbA1c ($>$ 8.5% [69 mmol/mol] or $\leq$ 8.5% [69 mmol/mol]) as a stratification factor. Descriptive statistical analyses were used to summarize secondary efficacy endpoints at Week 52 using an observed cases approach. Statistical analyses were performed using SAS 9.4 software.

3 | RESULTS

3.1 | Patient disposition and baseline characteristics

Patients were enrolled between November 13, 2017 and February 1, 2019, and the cut-off date was May 13, 2020. All the 495 randomized patients received at least one dose of assigned treatment, and 437 (88.3%; 86.7% to 89.1% across groups) completed the 24-week treatment period. Of the 427 patients who entered the extension period and received study treatment, 398 patients

(93.2%; 91.6%–98.5% across groups) completed the preplanned 52-week treatment period (Figure 1). In the 24-week period, 483 patients were included in the FAS and 490 in the SAS; in the 28-week extension period, 423 were included in the FAS and 427 in the SAS. The most common reason for treatment discontinuation was patient withdrawal during both 24-week placebo-controlled period (22/483; 4.6%) or 28-week extension period (8/75; 1.9%). There were 27 patients (5.6%; 25 [15.5%] in the placebo group) in the 24-week period and eight patients (2.0%) in the 28-week extension period who received rescue medication, including glimepiride, gliclazide and sitagliptin.

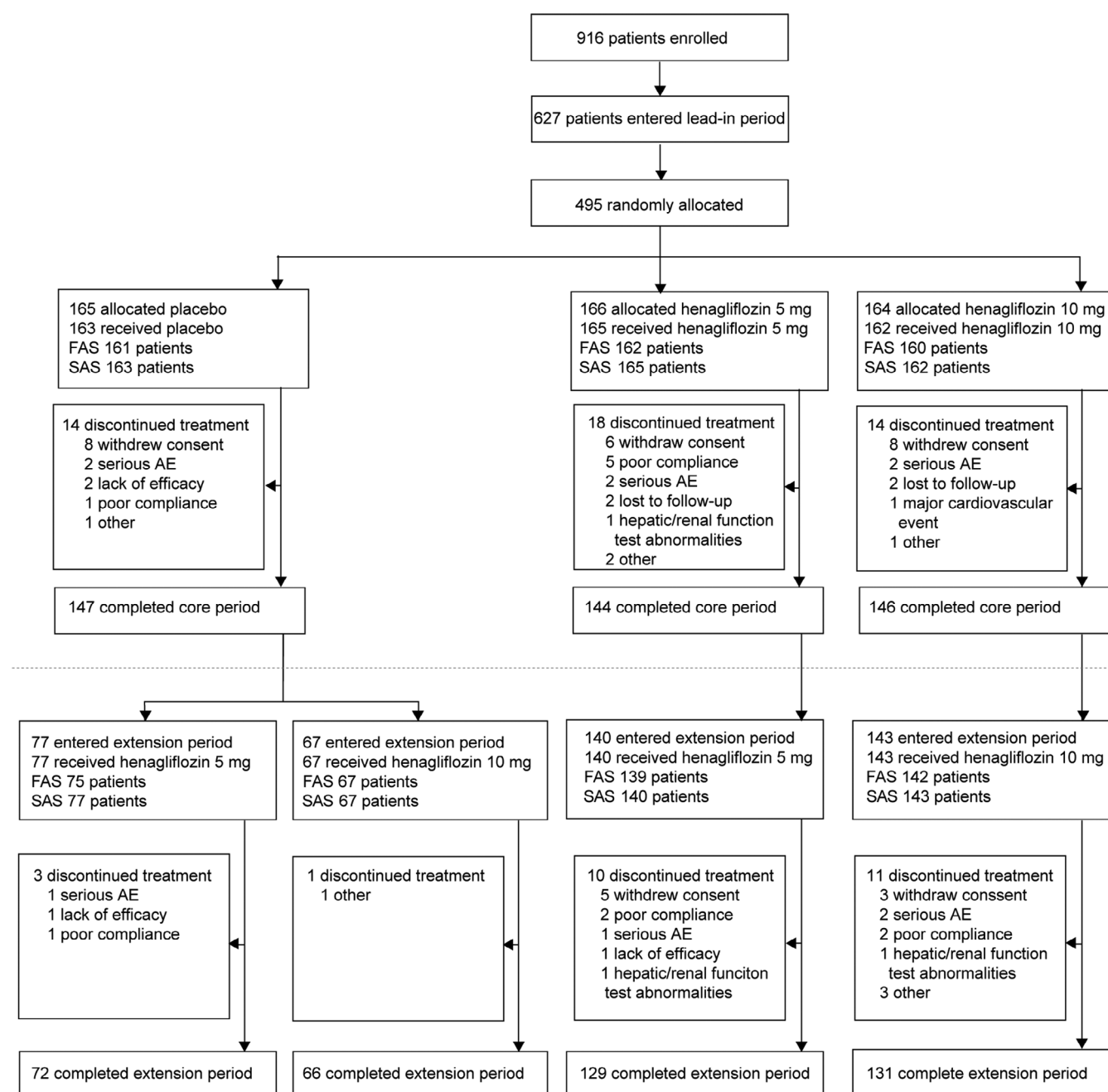


FIGURE 1 Trial profile. Abbreviations: AE, adverse event; FAS, full analysis set; SAS, safety set

TABLE 1 Demographics and baseline characteristics in the full analysis set

	Placebo (n = 161)	Henagliflozin 5 mg (n = 162)	Henagliflozin 10 mg (n = 160)
Age, years	55.3 ± 9.5	54.3 ± 9.5	54.7 ± 10.7
Men, n (%)	93 (57.8)	103 (63.6)	101 (63.1)
Body weight, kg	68.9 ± 11.2	69.8 ± 11.7	70.2 ± 12.0
BMI, kg/m ²	25.4 ± 3.1	25.5 ± 2.9	25.6 ± 3.2
Duration of T2DM, months	78.9 ± 69.7	66.4 ± 59.6	76.7 ± 57.6
HbA1c, mmol/mol	69 ± 9.8	69 ± 9.8	68 ± 9.8
HbA1c, %	8.5 ± 0.9	8.5 ± 0.8	8.4 ± 0.9
FPG, mmol/L	9.8 ± 2.0	10.0 ± 2.0	10.0 ± 2.0
2-hour PPG, mmol/L	15.5 ± 3.4	15.7 ± 3.1	15.6 ± 3.0
PPG AUC _{0-2 h} , mmol×h/L	28.7 ± 5.3	29.0 ± 4.8	29.0 ± 5.1
LDL cholesterol, mmol/L	2.8 ± 0.9	2.9 ± 0.8	2.9 ± 0.8
eGFR, mL/min/1.73m ²	128.0 ± 34.1	125.4 ± 31.3	128.9 ± 32.6
SBP, mmHg	127.4 ± 13.6	126.8 ± 13.7	127.3 ± 12.1
DBP, mmHg	79.1 ± 9.3	78.3 ± 9.6	79.2 ± 8.7

Abbreviations: AUC, area under the curve; BMI, body mass index; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; FPG, fasting plasma glucose; HbA1c, glycated haemoglobin; LDL, low-density lipoprotein; PPG, postprandial plasma glucose; SBP, systolic blood pressure; T2DM, type 2 diabetes mellitus.

Note: Continuous data are mean ± standard deviation.

Baseline demographics and disease characteristics were balanced across all groups (Table 1). The mean patient age was 54.8 years, and 297 patients (61.5%) were male. The mean baseline HbA1c was 8.4% (68 mmol/mol), and the median duration of T2DM was 58.3 months.

3.2 | Efficacy

At Week 24, a significant improvement in the change from baseline in HbA1c was observed for both the **henagliflozin 5-mg (LS mean difference: −0.76% [95% CI −0.93 to −0.60]/−8.3 mmol/mol [95% CI −10.2 to −6.6]; $P < 0.0001$)** and the **henagliflozin 10-mg groups (LS mean difference: −0.80% [95% CI −0.97 to −0.63]/−8.7 mmol/mol [95% CI −10.6 to −6.9]; $P < 0.0001$)** compared with the placebo group (Table 2). Furthermore, sensitivity analyses demonstrated that LS mean reductions from baseline in HbA1c remained greater with both doses of henagliflozin groups compared with placebo, regardless of whether values after initiation of rescue therapy were excluded or not (**Table S1 in Appendix S1**). Further subgroup analysis showed that the LS mean reductions in HbA1c at Week 24 were consistently more pronounced in patients receiving henagliflozin (5 or 10 mg) than in patients receiving placebo across all prespecified subgroups (Table S2). For patients with a baseline HbA1c ≤8.5% [69 mmol/mol], the placebo-adjusted LS mean changes in HbA1c at Week 24 for the henagliflozin 5- and 10-mg groups were −0.68% (95% CI −0.88 to −0.48)/−7.4 mmol/mol (95% CI −9.6 to −5.2) and −0.69% (95% CI −0.89 to −0.49)/−7.5 mmol/mol (95% CI −9.7 to −5.4), respectively (all $P < 0.0001$). **For patients with a baseline HbA1c >8.5% [69 mmol/mol], the placebo-adjusted LS mean changes in HbA1c at Week 24 for the henagliflozin 5- and 10-mg groups were −0.88% (95% CI −1.19 to −0.58)/−9.6**

mmol/mol (95% CI −13.0 to −6.3) and −0.97% (95% CI −1.27 to −0.66)/−10.6 mmol/mol (95% CI −13.9 to −7.2), respectively (all $P < 0.0001$). Similarly, other secondary glycaemic variables at Week 24, including reductions from baseline in FPG (Figure S2B), 2-hour PPG and PPG AUC_{0-2h}, as well as proportion of patients achieving HbA1c <7.0% (53 mmol/mol), were all more favourable in the henagliflozin 5-mg or 10-mg groups compared with the placebo group (all $P < 0.0001$; Table 2).

At Week 24, obvious reductions in body weight were noted in both the henagliflozin 5-mg group (LS mean difference −1.25 kg [95% CI −1.76 to −0.75]; $P < 0.0001$) and the henagliflozin 10-mg group (LS mean difference −1.22 kg [95% CI −1.72 to −0.72]; $P < 0.0001$) compared with the placebo group (Table 2). Additionally, impressive reductions in either SBP or DBP were observed for both henagliflozin dose groups compared with the placebo group at Week 24 (Table 2). The placebo-adjusted LS mean changes in SBP were −4.6 mmHg (95% CI −7.0 to −2.2) and −6.6 mmHg (95% CI −8.9 to −4.2) for the henagliflozin 5-mg group and the henagliflozin 10-mg group, respectively (all $P < 0.0001$); and the placebo-adjusted LS mean changes in DBP were −2.0 mmHg (95% CI −3.6 to −0.4) and −2.8 mmHg (95% CI −4.4 to −1.3), respectively (all $P < 0.05$).

The results for secondary efficacy endpoints **at Week 52** are provided in Table S3. **The mean change at Week 52 from baseline in HbA1c was −1.11 ± 0.82% (−12.1 ± 9.0 mmol/mol) for the placebo-henagliflozin 5-mg group and −0.94 ± 0.76% (−10.3 ± 8.3 mmol/mol) for the placebo-henagliflozin 10-mg group;** in patients in the henagliflozin 5-mg group and the henagliflozin 10-mg group, the mean changes were −1.06 ± 0.74% (−11.6 ± 8.1 mmol) and −1.07 ± 0.82% (−11.7 ± 9.0 mmol), respectively (Table S3 and Figure 2). Similarly, the reductions in FPG, 2-hour PPG and PPG

TABLE 2 Primary endpoint and secondary efficacy endpoints at week 24 in the full analysis set

Endpoints	Placebo (n = 161)	Henagliflozin 5 mg (n = 162)	Henagliflozin 10 mg (n = 160)
HbA1c			
n ^a	159	162	160
Change from baseline, LS mean difference (95% CI), mmol/mol	−2.6 (−3.9, −1.3)	−10.9 (−12.2, −9.6)	−11.4 (−12.7, −10.1)
Change from baseline, LS mean difference (95% CI), %	−0.24 (−0.36, −0.12)	−1.00 (−1.12, −0.88)	−1.04 (−1.16, −0.92)
Change vs. placebo, LS mean difference (95% CI), mmol/mol	–	−8.3 (−10.2, −6.6)	−8.7 (−10.6, −6.9)
Change vs. placebo, LS mean difference (95% CI), %	–	−0.76 (−0.93, −0.60)	−0.80 (−0.97, −0.63)
P value ^b		<.0001	<.0001
FPG (mmol/L)			
n ^a	159	162	160
Change from baseline, LS mean difference (95% CI)	−0.34 (−0.60, −0.07)	−2.06 (−2.32, −1.80)	−2.23 (−2.49, −1.97)
Change vs. placebo, LS mean difference (95% CI)	–	−1.72 (−2.09, −1.35)	−1.89 (−2.26, −1.52)
P value ^b		<.0001	<.0001
2-hour PPG (mmol/L)			
n ^c	128	143	145
Change from baseline, LS mean difference (95% CI)	−1.19 (−1.62, −0.76)	−4.05 (−4.45, −3.64)	−3.92 (−4.32, −3.52)
Change vs. placebo, LS mean difference (95% CI)	–	−2.85 (−3.44, −2.26)	−2.73 (−3.32, −2.14)
P value ^b		<.0001	<.0001
PPG AUC_{0-2 h} (mmolxh/L)			
n ^c	128	142	145
Change from baseline, LS mean difference (95% CI)	−2.04 (−2.71, −1.37)	−6.37 (−7.00, −5.74)	−6.42 (−7.05, −5.79)
Change vs. placebo, LS mean difference (95% CI)	–	−4.33 (−5.25, −3.40)	−4.37 (−5.30, −3.45)
P value ^b		<.0001	<.0001
Body weight (kg)			
n ^a	160	162	160
Change from baseline, LS mean difference (95% CI)	−1.39 (−1.74, −1.03)	−2.64 (−3.00, −2.29)	−2.61 (−2.96, −2.25)
Change vs. placebo, LS mean difference (95% CI)		−1.25 (−1.76, −0.75)	−1.22 (−1.72, −0.72)
P value ^b		<.0001	<.0001
SBP (mmHg)			
n ^a	160	162	160
Change from baseline	−2.0 (−3.6, −0.3)	−6.5 (−8.2, −4.9)	−8.5 (−10.2, −6.8)
Change vs. placebo, LS mean difference (95% CI)	–	−4.6 (−7.0, −2.2)	−6.6 (−8.9, −4.2)
P value ^b		.0002	<.0001
DBP (mmHg)			
n ^a	160	162	160
Change from baseline, LS mean difference (95% CI)	−0.7 (−1.8, 0.4)	−2.7 (−3.8, −1.5)	−3.5 (−4.6, −2.4)
Change vs. placebo, LS mean difference (95% CI)	–	−2.0 (−3.6, −0.4)	−2.8 (−4.4, −1.3)
P value ^e	–	.0138	.0004
Patients with HbA_{1c} < 6.5% (48 mmol/mol)			
n/N ^d	6/161	13/162	12/160
Proportion of patients, % (95% CI) ^e	3.7 (1.4, 7.9)	8.0 (4.3, 13.3)	7.5 (3.9, 12.7)
P value ^f	–	.0896	.1400
Patients with HbA_{1c} < 7.0%			
n/N ^d	18/161	52/162	52/160

TABLE 2 (Continued)

Endpoints	Placebo (n = 161)	Henagliflozin 5 mg (n = 162)	Henagliflozin 10 mg (n = 160)
Proportion of patients, % (95% CI) ^e	11.2 (6.8, 17.1)	32.1 (25.0, 39.9)	32.5 (25.3, 40.3)
P value ^f	–	<.0001	<.0001

Abbreviations: AUC, area under curve; CI, confidence interval; DBP, diastolic blood pressure; FPG, fasting plasma glucose; HbA1c, glycated haemoglobin; LS, least squares; PPG, postprandial glucose; SBP, systolic blood pressure.

^aNumber of analysed patients using the last observation carried forward (LOCF) approach.

^bHenagliflozin versus placebo were compared using the ANCOVA model excluding values after initiation of rescue therapy.

^cNumber of analysed patients using an observed cases approach.

^dNumber of patients achieving HbA1c <6.5% (48 mmol/mol) or HbA1c <7.0% (53 mmol/mol) at Week 24 analysed with nonresponder imputation (patients with missing HbA1c value at Week 24 were considered as nonresponders) / number of randomized patients with Week 24 (LOCF) values.

^e95% CIs were calculated using the Clopper-Pearson method.

^fP value for categorical variables compared between treatment group and placebo were from Cochran-Mantel-Haenszel tests for general association with baseline HbA1c (HbA1c >8.5% [69 mmol/mol] or HbA1c ≤8.5% [69 mmol/mol]) as a stratification factor.

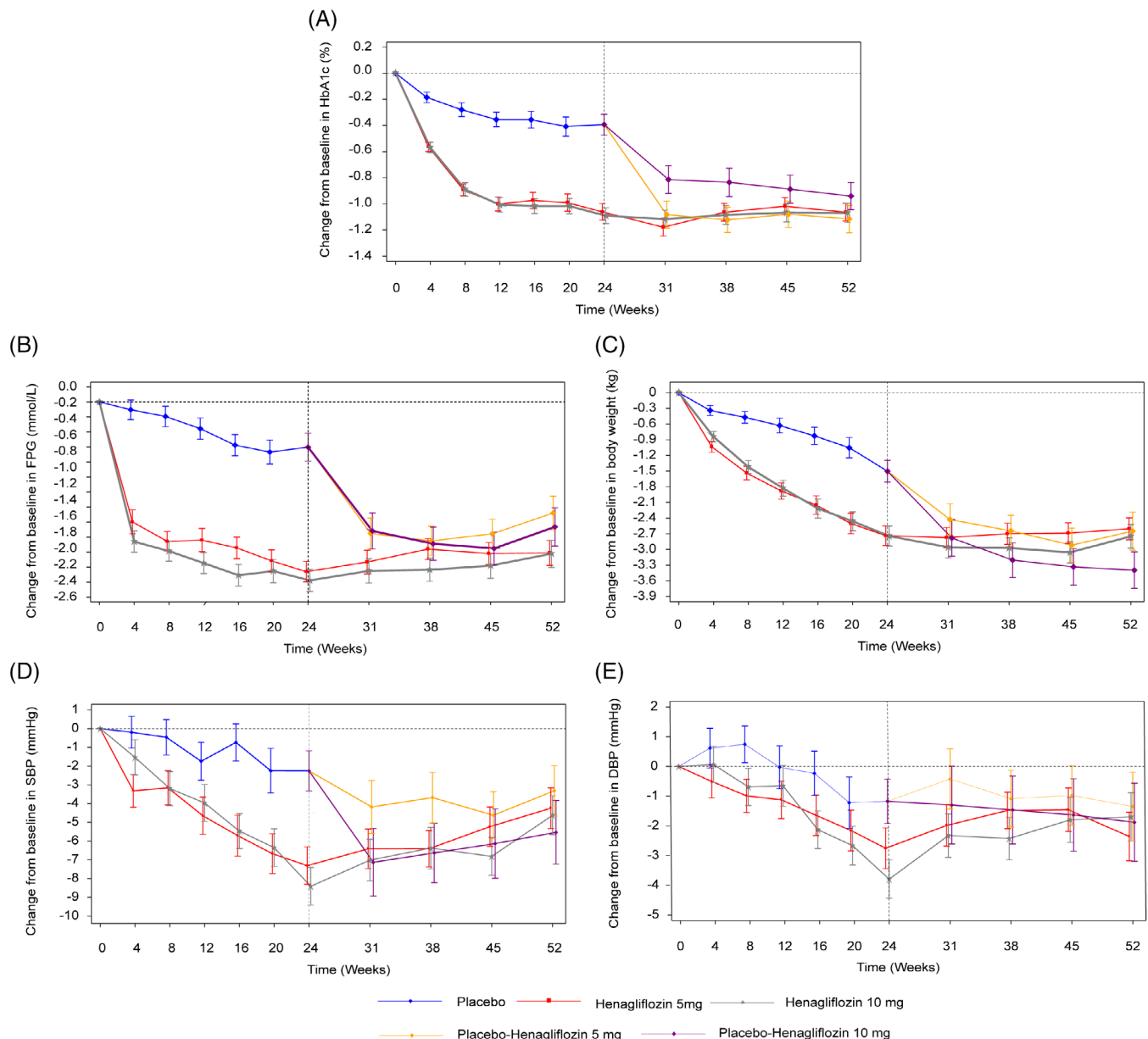


FIGURE 2 Observed change from baseline in metabolic variables and blood pressure over the 52 weeks. **A**, glycated haemoglobin (HbA1c). **B**, Fasting plasma glucose (FPG). **C**, Body weight. **D**, systolic blood pressure (SBP). **E**, Diastolic blood pressure (DBP). Data are mean ± standard error

AUC_{0-2h}, body weight and BP at Week 24 were sustained at Week 52 in patients who continued henagliflozin 5- or 10-mg treatment during the extension period. Additionally, these variables were all

reduced compared to the respective baseline values in those who switched to henagliflozin during the extension period (Table S3 and Figure 2).

TABLE 3 Adverse events during 24-week treatment period in the safety set

	Placebo (n = 163)	Henagliflozin 5 mg (n = 165)	Henagliflozin 10 mg (n = 162)
AE	115 (70.6)	132 (80.0)	137 (84.6)
Treatment-related AE	38 (23.3)	49 (29.7)	59 (36.4)
AE leading to study discontinuation	2 (1.2)	4 (2.4)	4 (2.5)
Serious AE	7 (4.3)	13 (7.9)	5 (3.1)
Serious AE leading to study discontinuation	2 (1.2)	3 (1.8)	3 (1.9)
Deaths	0	1 (0.6)	0
Events of special interest			
Hypoglycaemia	2 (1.2)	4 (2.4)	5 (3.1)
Major episode of hypoglycaemia ^a	0	0	0
Hypotension	0	1 (0.6)	0
UTI ^b	10 (6.1)	5 (3.0)	5 (3.1)
Genital infection ^c			
Male	0	0	0
Female	0	3 (2.0)	1 (0.6)
AEs occurring in ≥2% patients in either group and more frequently with henagliflozin ^d			
Urine ketone body presence	14 (8.6)	24 (14.5)	26 (16.0)
Hyperlipidaemia ^e	14 (8.6)	19 (11.5)	16 (9.9)
Ketosis ^f	2 (1.2)	5 (3.0)	16 (9.9)
Urine ACR increased	6 (3.7)	9 (5.5)	12 (7.4)
Weight decreased ^g	3 (1.8)	9 (5.5)	11 (6.8)
Blood PTH increased	4 (2.5)	11 (6.7)	9 (5.6)
Toothache	1 (0.6)	8 (4.8)	6 (3.7)
Blood ketone bodies increased	1 (0.6)	7 (4.2)	6 (3.7)
Protein urine present	2 (1.2)	5 (3.0)	6 (3.7)
Hepatic steatosis	1 (0.6)	5 (3.0)	5 (3.1)
Haemoglobin increased	0	3 (1.8)	4 (2.5)
Specific gravity urine increased	0	0	4 (2.5)
Hypertension	4 (2.5)	8 (4.8)	3 (1.9)
Nephrolithiasis	2 (1.2)	6 (3.6)	2 (1.2)
Bronchitis	1 (0.6)	6 (3.6)	2 (1.2)
Benign prostatic hyperplasia	0	4 (2.4)	2 (1.2)
Electrocardiogram ST segment abnormal	0	4 (2.4)	1 (0.6)
Herpes zoster	0	4 (2.4)	1 (0.6)

Abbreviations: AE, adverse event; ACR, albumin/creatinine ratio; PTH, parathyroid hormone; UTI, urinary tract infection.

Note: Data are n (%).

^aA major episode of hypoglycaemia was defined as symptomatic episodes requiring third party assistance and/or with a capillary or plasma glucose less than 2.8 mmol/L.

^bEvents were UTI or asymptomatic bacteriuria.

^cEvents were genital infection, vulvitis or vaginal infection.

^dAll AEs occurring more frequently in the henagliflozin groups (either dose group) with a difference of ≥2% versus placebo were listed.

^eHyperlipidaemia defined as elevated serum total cholesterol and/or triglyceride, or reduced high-density lipoprotein.

^fKetosis defined as increased blood and/or urine ketone levels.

^gPatients experienced obvious weight loss per investigator's judgement during a short period, "weight decreased" coded by the Medical Dictionary for Regulatory Activities version 22.0 preferred term was recorded.

3.3 | Safety

Safety data throughout the 24-week treatment period are provided in Table 3. Generally, the incidence of AEs was balanced across all treatment groups, ranging from 70.6% to 84.6%. The average treatment compliance was high for all three treatment groups, ranging from 98.3% to 98.9%. Treatment discontinuations due to AEs were few, with 2.4%, 2.5% and 1.2% reported for the henagliflozin 5-mg, henagliflozin 10-mg and placebo groups, respectively. Serious AEs were reported for seven patients (4.3%) in the placebo group, 13 (7.9%) in the henagliflozin 5-mg group and five (3.1%) in the henagliflozin 10-mg group (Table S4). Treatment-related serious AEs occurred in only two patients (1.2%) in the henagliflozin 5-mg group ($n = 1$, ketosis and UTI; $n = 1$, hypotension), and all of them were cured or improved after active treatment. One patient in the henagliflozin 5-mg group (a 65-year-old man) died from primary hepatocellular carcinoma with bone metastasis, which was not associated with the study treatment in the investigator's judgement.

During the 24-week placebo-controlled period, AEs occurring more frequently in the henagliflozin groups versus the placebo group mainly included ketosis-related events and transient laboratory test abnormalities (Table 3). Genital infection was reported in three patients (1.8%) in the henagliflozin 5-mg group and one patient (0.6%) in the henagliflozin 10-mg group, and all were women. The frequency rates of hypoglycaemia (2.4%, 3.1% and 1.2%, respectively) and UTI (3.0%, 3.1% and 6.1%, respectively) were similar in the henagliflozin 5-mg, henagliflozin 10-mg and placebo groups. **No cases of diabetic ketoacidosis or major episodes of hypoglycaemia were reported,** and only one patient (in the henagliflozin 5-mg group) experienced hypotension throughout 24 weeks. The complete list of AEs occurring in $\geq 2\%$ patients in either group up to Week 24 are presented in Table S5.

The safety data regarding the 28-week extension period are presented in Table S6. Overall, the main type, frequency and severity of AEs in patients on henagliflozin (continued or switched to) during the extension period were consistent with those reported in patients on henagliflozin during the initial 24-week treatment period.

4 | DISCUSSION

Henagliflozin is the first SGLT2 inhibitor independently developed in China, and this was the first phase 3 trial conducted in an all-Chinese T2DM population reporting the results obtained with henagliflozin as add-on therapy to metformin. In this phase 3 trial of patients with T2DM on background metformin, once-daily henagliflozin 5 or 10 mg clinically and significantly reduced HbA1c from baseline, and improved body weight and blood pressure after a 24-week placebo-controlled treatment period; these benefits were sustained throughout additional 28-week extension period. Additionally, henagliflozin had a good tolerability profile, with high medication compliance over the 52-week treatment.

With both doses of henagliflozin, the magnitude of reduction in HbA1c from baseline at Week 24 in T2DM patients with inadequate glycaemic control with metformin monotherapy was clinically relevant, which is consistent with reports on other SGLT2 inhibitors as add-on to metformin.^{12,13} Additionally, **the glycaemic efficacy of henagliflozin was prominent across all prespecified subgroups, suggesting that a broad spectrum of patients could obtain clinical benefit.** Meanwhile, treatment with henagliflozin also brought about synchronous improvements in other glycaemic measures, including FPG, 2-hour PPG and PPG AUC_{0-2h}, and in the proportion of patients with HbA1c $<7.0\%$ (53 mmol/mol) at Week 24 relative to placebo, which were also competitive with findings from other SGLT2 inhibitors (Table S7), and these effects could be maintained over the 52-week study period.¹²⁻¹⁹

Unreasonable blood pressure control and obesity are widely recognized cardiovascular risk factors in the multifactorial management of T2DM patients.²⁰ In our phase 3 trial, 24-week treatment with henagliflozin resulted in a clinically meaningful reduction in both SBP (5 and 10 mg: -6.5 and -8.5 mmHg, respectively) and DBP (5 and 10 mg: -2.7 and -3.5 mmHg, respectively) versus placebo, with only one patient in the henagliflozin 5-mg group experiencing a hypotension episode. The antihypertensive mechanisms of SGLT2 inhibitors are not entirely understood, but could involve increased urinary sodium excretion and improved vascular endothelial function.¹ The underlying mechanism for the antihypertensive effects observed for henagliflozin requires further investigation. Emerging evidence (EMPA-REG OUTCOME trial, CANVAS Program, DECLARE-TIMI 58) suggests that SGLT2 inhibitors might be effective in the prevention of heart failure in T2DM patients,²¹⁻²³ which could be partly explained by the blood pressure reduction effects. Our results also confirmed that henagliflozin as add-on therapy to metformin was associated with clinically relevant body weight reduction from baseline (2.61-2.64 kg for henagliflozin at Week 24; 2.60-2.75 kg for henagliflozin at Week 52) in an all-Chinese population characterized by lower body weight and higher proportion of visceral fat mass compared with White populations.²⁴ **The weight-lowering effects** of SGLT2 inhibitors have been demonstrated to be accompanied by reductions in total body fat mass, **which is mainly attributed to a reduction in visceral adipose tissue and ectopic (liver) fat instead of subcutaneous adipose tissue.**^{5,25-27} The encouraging and durable blood pressure or body weight reductions of henagliflozin were not just accidental phenomena; similar results were also observed in a separate phase 3 trial (NCT03159052) involving T2DM patients inadequately controlled on diet and exercise receiving henagliflozin monotherapy. Our study demonstrates that the improvement in body weight and blood pressure with henagliflozin was sustainable, and the long-term benefits of potential reduction of cardiovascular events should be investigated in the near future, especially in high-risk populations.

Both doses of henagliflozin were generally safe and well tolerated during the 52-week study period. The incidence of ketosis, presence of urine ketone bodies and increased blood ketone bodies were slightly more frequent in the henagliflozin groups compared with the placebo group, and this might be expected from the shift from elevated urine glucose excretion to lipids and ketone bodies as a source

of energy fuel.²⁸ Importantly, the majority of ketosis-related events were mild, which was in line with the low occurrence rate of diabetic ketoacidosis documented with other SGLT2 inhibitors.²⁹ The incidence of UTI was not higher in either of the henagliflozin groups compared with the placebo group, which was consistent with results from a meta-analysis of 86 randomized controlled trials,³⁰ reporting that the risk of UTI was not increased with SGLT2 inhibitor treatment compared with placebo or active control. Remarkably, the incidence of genital infections, a common AE reported with other SGLT2 inhibitors (Table S7), was relatively low (<2%) across all treatment groups. Hypoglycaemia events were infrequent and similar among the henagliflozin 5-mg, henagliflozin 10-mg and placebo groups, also suggesting that metformin in combination with henagliflozin did not confer a high risk of hypoglycaemia.³¹

The present study has several limitations. First, a study duration beyond 52 weeks warrants further exploration of the long-term efficacy and durable safety of henagliflozin, because T2DM is a chronic disease. Second, the findings during the 28-week extension period do not allow conclusions to be drawn regarding the net effects of henagliflozin because of the lack of a concurrent placebo control arm. Third, future studies on the comparative efficacy (glycaemic and non-glycaemic, especially cardiovascular outcomes and renal protections) and safety of henagliflozin versus other OADs, such as dipeptidyl peptidase-4 inhibitors, as add-on to metformin should also be explored in active-controlled trials in future.

In conclusion, henagliflozin (5 or 10 mg) as add-on therapy to metformin improved glycaemic control, reduced body weight and blood pressure, and was generally well tolerated in Chinese patients with T2DM who were inadequately controlled with metformin monotherapy.

ACKNOWLEDGMENTS

This study was funded by Jiangsu Hengrui Medicine. We are grateful to all the patients and their families and all members of the study group. Medical writing support was provided by Lin Dong, PhD (a medical writer at Jiangsu Hengrui Medicine) according to Good Publication Practice Guidelines.

CONFLICTS OF INTEREST

H.L., T.Z. and J.Z. are employees of Jiangsu Hengrui Medicine. All other co-authors declare no competing interests.

AUTHOR CONTRIBUTIONS

Author contributions were as follows. Design: J.W., H.L., T.Z. and J.Z. Conduct/data collection: J.W., L.Z., Y.Z., S.Q., X.W., P.L., L.F., B.M., S.Y., J.S., W.L., Z.L., D.C. and Z.C. Statistical analysis: T.Z. Interpretation of data: all authors. Drafting of the manuscript: J.W. Critical revision of the manuscript for important intellectual content: all authors. Approval of the final manuscript: all authors.

DATA AVAILABILITY STATEMENT

All data analyzed during this study are included in this published article or as supplementary information files.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

How to cite this article: Weng J, Zeng L, Zhang Y, et al. Henagliflozin as add-on therapy to metformin in patients with type 2 diabetes inadequately controlled with metformin: A multicentre, randomized, double-blind, placebo-controlled, phase 3 trial. *Diabetes Obes Metab*. 2021;1-11. <https://doi.org/10.1111/dom.14389>